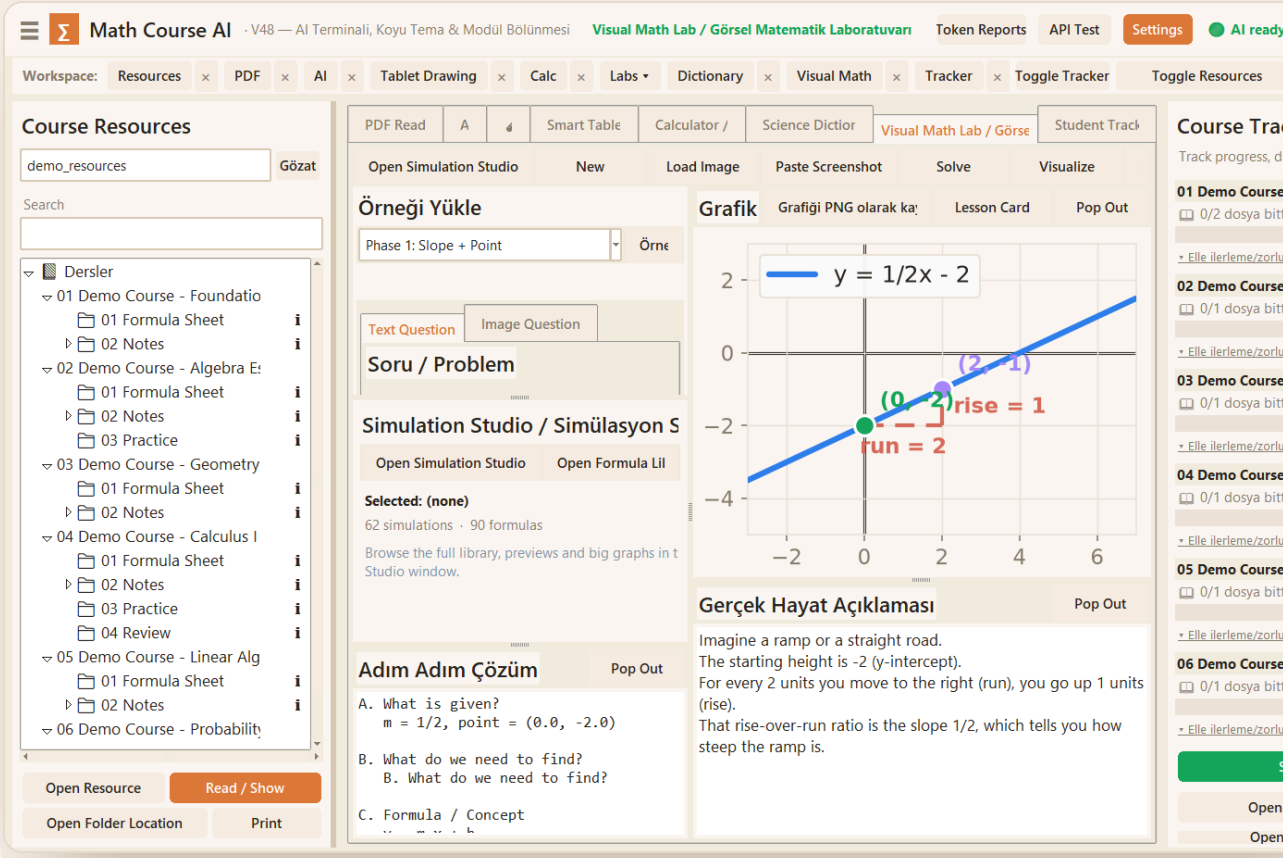
Math Course AI

Everything you need while following a maths course — in one window.

Read and annotate your PDFs, clip questions into your own bank, study with spaced repetition — and when you get stuck, ask an AI that runs on your own computer.

v1.0.0 · Windows 10/11 (x64) · macOS (Apple Silicon) · Türkçe · English · Open source · MIT



Course work is scattered. Math Course AI gathers it up.

	THE PROBLEM	WHAT MATH COURSE AI DOES
01	*“The PDF is in one place, notes in another, questions in a notebook.”*	**One window, one database.** Your material, notes and questions live together, and the program remembers where you left off.
02	*“I don't know what to study today.”*	**The Today dashboard** shows what is due, what you got wrong and what you never tried; one click starts an exam on it.
03	*“I forget what I learned.”*	**Spaced repetition:** a question you get right returns after 1 → 3 → 7 → 14 → 30 → 60 days; one you get wrong comes back at once.
04	*“There is nobody to ask when I'm stuck.”*	**A local AI tutor** (LM Studio) and the **Visual Math Lab** with step-by-step solutions.
05	*“I can't find practice questions.”*	**Built-in generators** make unlimited, licence-free, bilingual questions; or take questions straight from a PDF with **Clip Question**.

One application, a complete study desk

No account to create and no server to run. Your study data lives in a local database on your own computer.

13
workspaces
PDF, AI, exams, labs, dictionary...

5
science labs
Data, Physics, Chemistry, DNA, Quantum

665
topic notes
open instantly, no AI involved

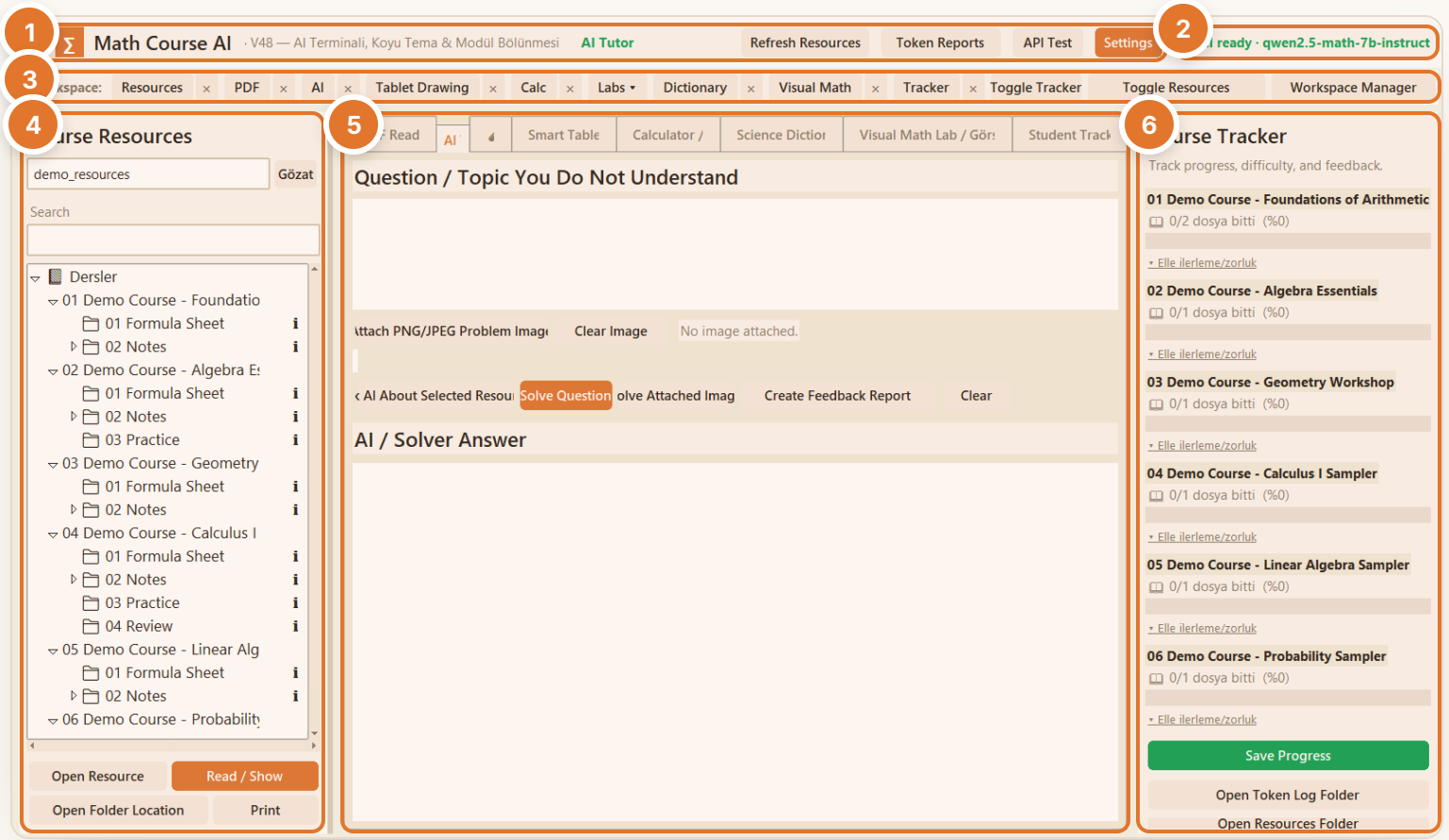
50
embedded simulations
interactive visuals of course topics

90
formula cards
show what a formula actually says

19
chapter user guide
inside the program, offline, TR and EN

None of these numbers is hand-written: each is measured from the source code by tools/measure_facts.py.

Meet the main window: three columns, one window



This and the following screens are real captures of the running program (synthetic demo course library, empty learner profile). Some panels still show Turkish labels because they were built before the language was switched; they come up in English after a restart.

1 Top bar
Menu, the name of the open workspace, Refresh Resources, Token Reports, API Test and Settings.

2 AI status light
Green: ready (with the model's name) · amber: no model loaded · red: off.

3 Workspace strip
Switch tools in one click; hide the ones you don't use with ×.

4 Course Resources
The tree of your own course folder, search and open/read buttons.

5 The workspace
PDF, AI, exams, labs… the tool you pick opens here.

6 Course Tracker
Files finished and progress per course.

05 — WORKFLOW

A typical study session in five steps

The program is built around following a course end to end: read → collect → review → measure.

1

Add your material

Put your course PDFs into the Resources folder, or pick your own folder with Browse.

Course Resources

→

2

Read and annotate

Open a PDF; draw, highlight, type. Your page is remembered.

PDF Reader

→

3

Bank the question

Drag a rectangle with Clip Question; it is saved with course, topic and difficulty.

PDF Reader → Clip

→

4

Practise

Start an exam; spaced repetition plans what comes back and when.

Student Tracker → Exam

→

5

See your progress

The Today dashboard, statistics and a printable weekly summary.

Student Tracker → Today

Whenever you get stuck: AI tutor · Visual Math · Dictionary · Calculator · Drawing board · Topic notes

Your own course folder, as a tidy tree



1. **Search**
   The tree filters as you type, so even in a big course your topic is right there.

2. **Lesson tree**
   Everything sits under two roots: Lessons and Question Banks. Tick a file when you finish it; opening a PDF marks it “Reading” automatically.

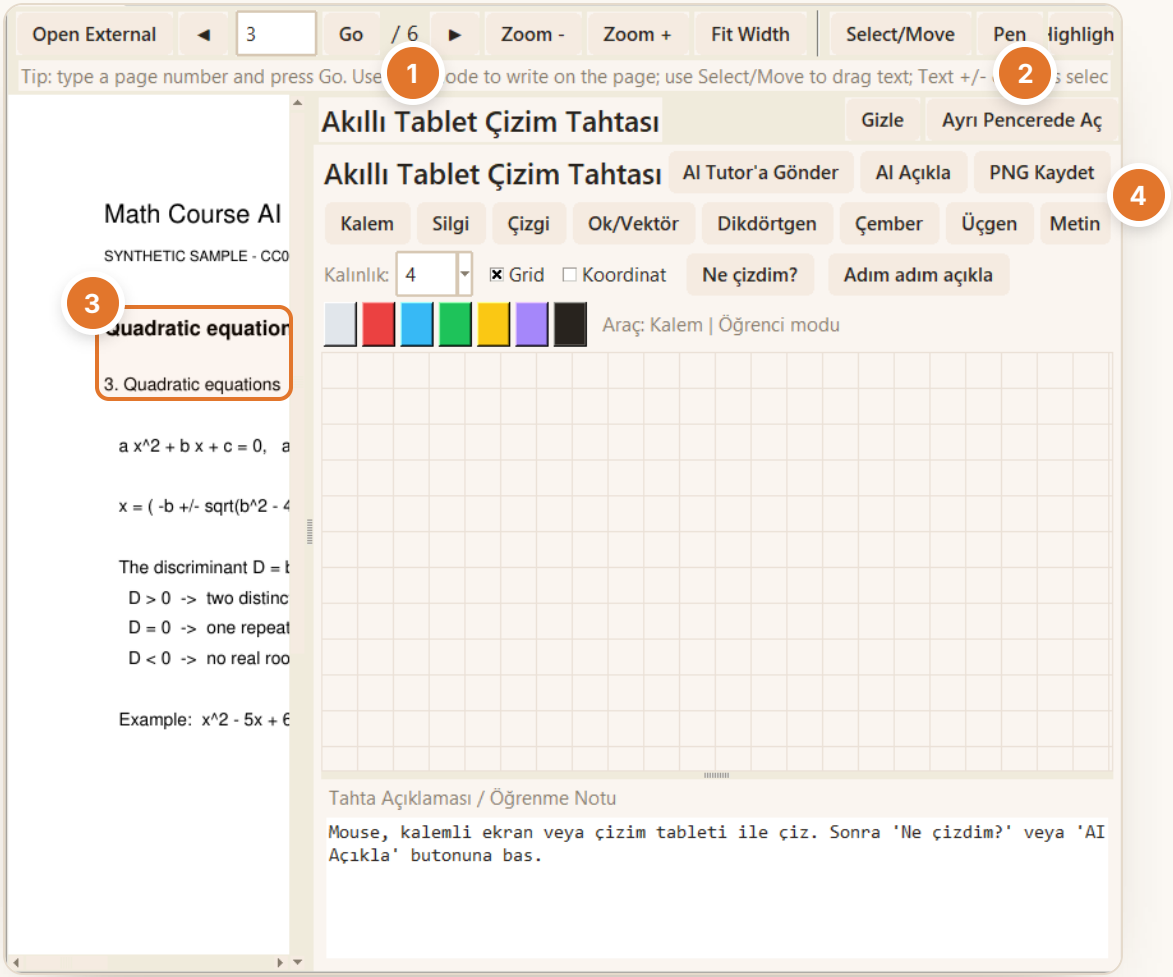
3. **Topic notes (i)**
   Instant notes for 665 topics with no AI: an everyday analogy → what it is → what it is for → its history where there is any. (The notes are written in Turkish.)

4. **Open · Read/Show · Print**
   Open the selected file in the program, show its folder or print it.

5. **Course Tracker**
   How many files you finished in each course and the percentage, plus optional manual progress and difficulty notes.

6. **Save Progress**
   The program also remembers your last PDF, its page and zoom.

Read, annotate, clip the question



1 Navigation and zoom
Go to a page, previous/next, zoom in and out, Fit Width.

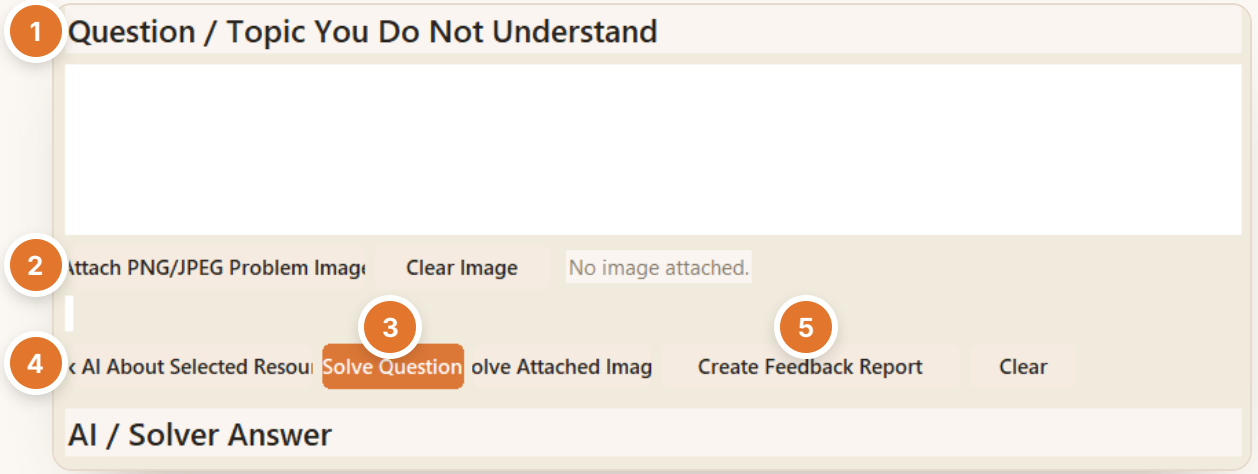
2 Annotation tools
Pen, highlighter, text and eraser. Notes go to a separate sidecar file — the original PDF is never modified.

3 Clip Question
Drag a rectangle around a question: it is cut at high resolution and enters your bank with course, topic, difficulty and choices. The source PDF and page are stored automatically.

4 A drawing board alongside
Work the solution out next to the page; hide it or pop it out.

You can export, print or share the annotated PDF.

Ask when you're stuck — the AI runs on your own computer





The status light in the top bar is genuinely polled every 30 seconds; click it for a connection test.

- **The default provider is local:** LM Studio runs on your own computer, and the model only receives the text and context you selected.
- **Optional:** without any AI the PDFs, notes, question bank, exams, labs and dictionary all work fully.

1. **Type your question**
   The problem or topic you don't understand.
2. **Attach an image**
   A photo or screenshot of the question: OCR is tried first, the vision model if needed.
3. **Solve Question**
   The answer appears in the “AI / Solver Answer” area below.
4. **Ask about the selected resource**
   Ask about the file you selected in the tree.
5. **Feedback report**
   Turns the question, the answer and your course progress into a report with next steps.
6. **Model profiles**
   Maths, vision, chemistry... each job is routed to a suitable model; removing a model lowers quality instead of breaking the feature.

Have code written — nothing runs until you approve it

Settings → AI Terminal: for sympy solutions, matplotlib plots and small helper scripts.

PROVIDER

Two providers

Local LM Studio is the default. **NVIDIA NIM** is optional: it is only used if you pick it in that session and accept the consent dialog; requests are logged on NVIDIA's side, and the window says so.

SAFETY

The approval gate

The complete code the AI wrote is shown in a separate window; nothing runs until you approve it. In this release code execution is **off by default**.

MEMORY

Work record

Every finished job goes into a permanent, searchable record: question, answer, model, duration — plus code and output if it ran. Export it as text or copy it.

AWARENESS

Token ledger

Which model did how much work: today, 7 days, 30 days; by model, provider and day. No request text is stored, only counters. Exports to CSV.

Shortcuts: `Ctrl+Enter` send `Ctrl+L` clear `Esc` stop

The API key is kept only in this program's own Windows Credential Manager entry; the screen shows its last 4 characters.

A ten-second answer to “what do I study today?”

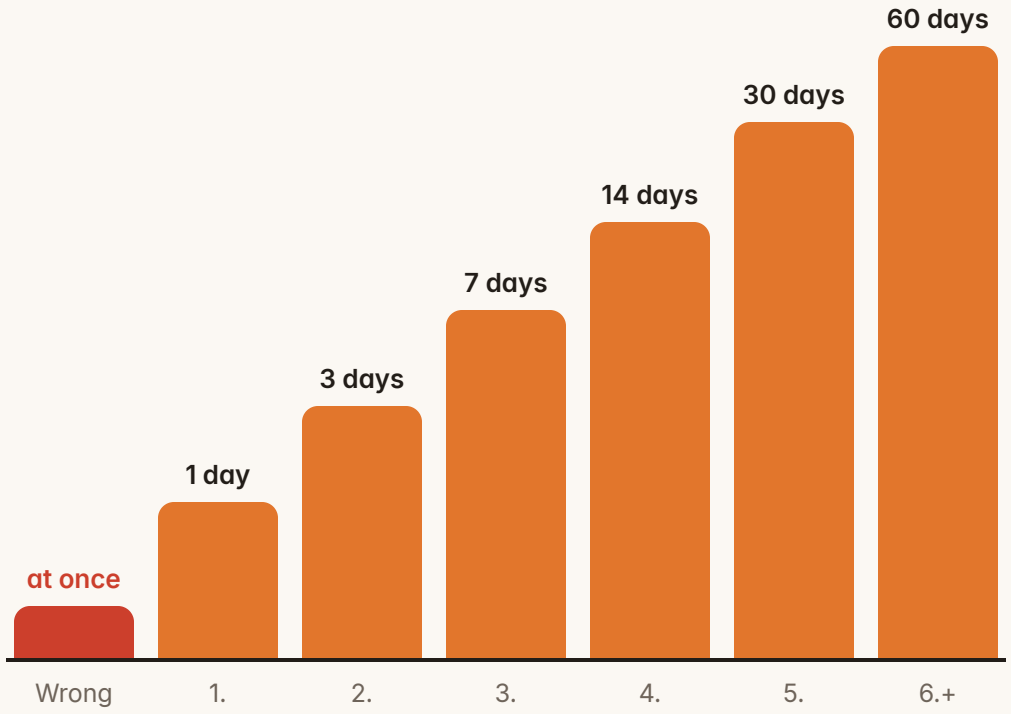


Captured on an empty demo profile: with no data the dashboard invents nothing — and says so.

1. **Six sub-tabs**
   Today · Learner · Question Bank · Question Packs · Exam · Statistics
2. **Four cards**
   Due for review, wrong answers, untried questions and accuracy.
3. **One-click exam**
   Each card's button starts an exam on that pool right away.
4. **This Week suggestions**
   Up to three reasoned suggestions from your data; rule-based, no AI. E.g. “Derivatives — 41.7% over the last 12 attempts”.
5. **Weak topics & progress**
   What to study first, and how far each course has got.

Question bank, exams and spaced repetition

When does a question come back?



Bottom axis: correct answers in a row. The date is computed from each question's own attempt history; the Today dashboard and the “Due” pool both read this queue.

- **Add questions** by hand, by bulk paste (question = answer), from CSV, or with Clip Question from a PDF.
- **Pick a pool:** All · Untried · Wrong · Review · Due.
- **Multiple-choice** is graded automatically. For **open answers** sympy recognises equivalence (0.5 = 1/2, 2x+2x = 4x) and suggests MATCH / EQUIVALENT / DIFFERENT — you make the final call.
- **Timed mode:** a countdown runs, red in the last minute; when time is up the exam closes itself.
- **Scratch pad:** work under every question with a mouse or pen.
- **Exam breakdown:** double-click a row afterwards — what you answered, the correct answer and the time spent, question by question.

Unlimited questions, shareable packs, a weekly report

QUESTION PACKS

Built-in question generators

Pre-Algebra, Algebra, Geometry, Linear Algebra and Calculus: each produces the question and its exact answer. Unlimited, licence-free, Turkish and English.

SHARING

Share with .mcpack

Export the questions you filtered, with licence and attribution, into one file; image questions you clipped travel too. A friend imports it in one click.

REPORT

Weekly summary report

One page in your browser: study days, questions answered, accuracy against last week, weak topics, finished exams. Saves as a colour or grayscale A4 PDF.

MEASUREMENT

Statistics

Overall accuracy, a per-topic mastery table, weak topics and the exam score trend. With no data it draws no empty chart — it says “no data this week”.

Pre-Algebra
arithmetic, fractions, percentages, ratios

Algebra
equations, quadratics, systems, exponents

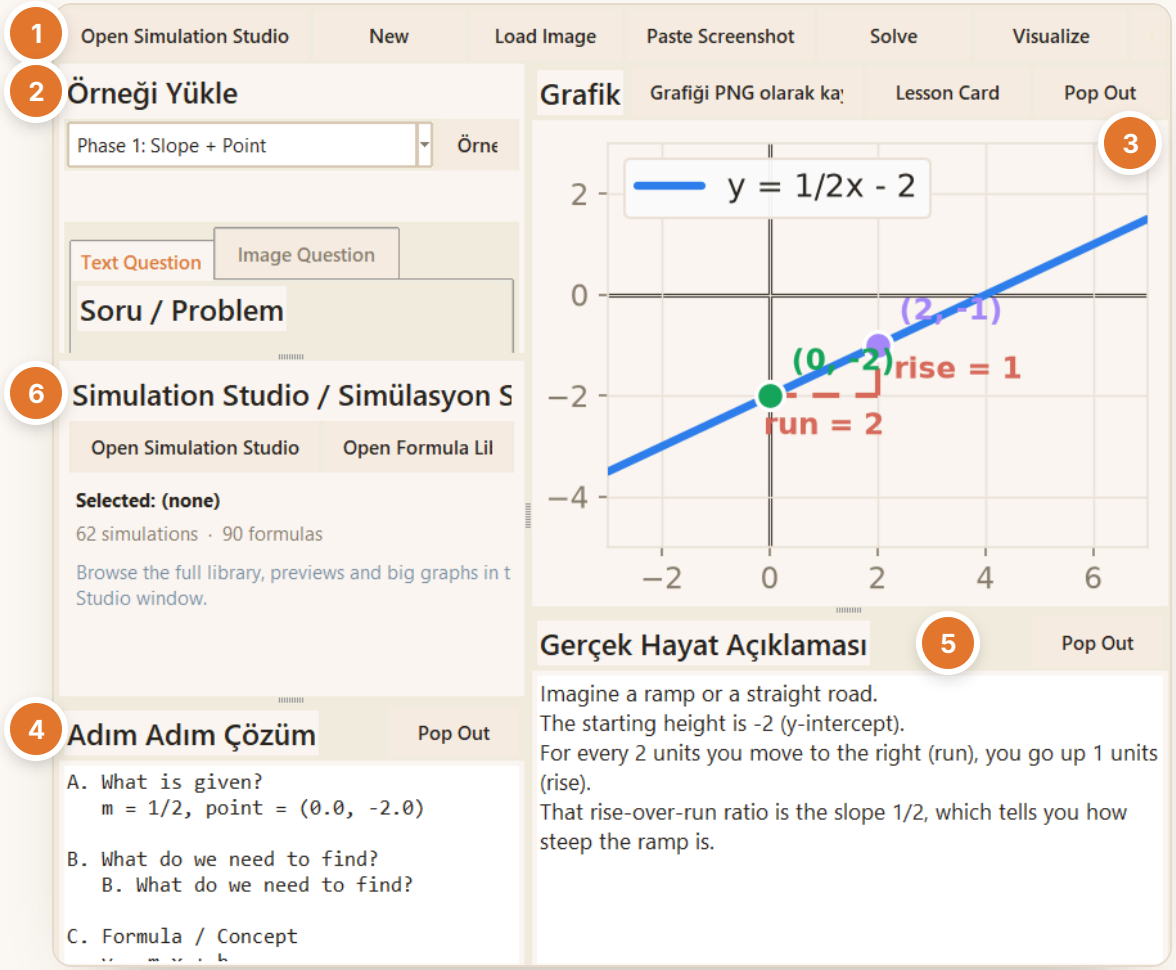
Geometry
area/perimeter, Pythagoras, circles

Linear Algebra
determinants, matrices, dot product

Calculus
slope, derivatives

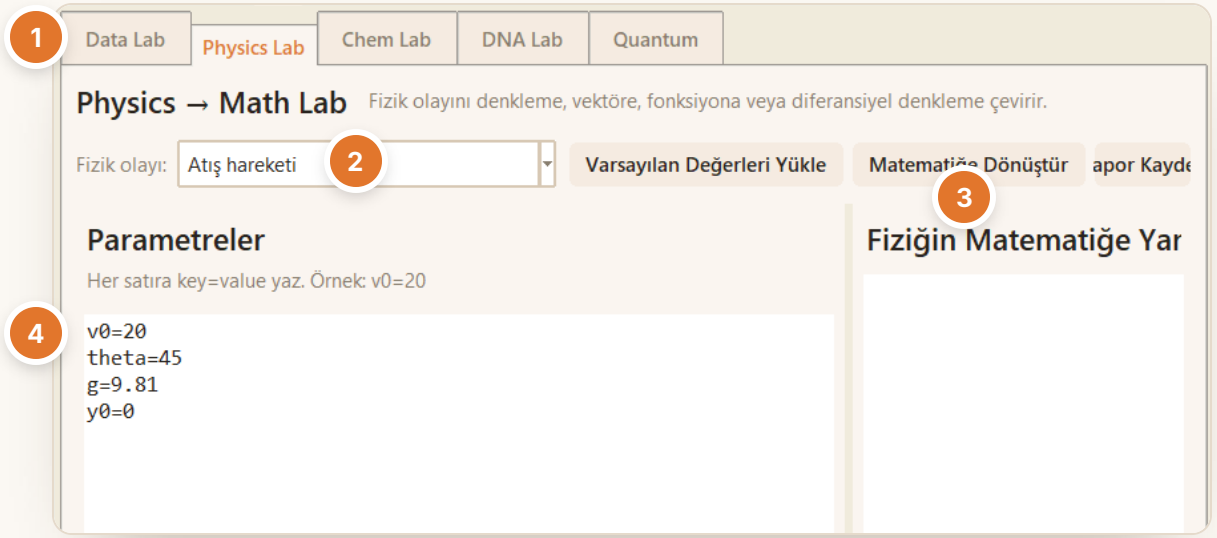
Licence rule: copyrighted question banks are never embedded. An external pack with no licence is marked UNKNOWN and warned about before import, and file paths inside a pack are validated.

See the path, not just the answer



1. **Toolbar**
   New, load an image, paste a screenshot, Solve, Visualize.
2. **Ready-made examples**
   Pick an example (e.g. slope + point) or type your own question / attach an image.
3. **Graph**
   A matplotlib plot; save it as PNG, make a lesson card, pop it out.
4. **Step-by-step solution**
   With sympy: what is given → what is asked → formula → working.
5. **Real-life explanation**
   The topic told through an everyday picture: slope = how steep a ramp is.
6. **Simulation Studio**
   50 embedded interactive simulations and 90 formula cards; ones built from your course folder are added to the list.

Five science labs — one idea: turn the subject into maths



1. **Five labs, one tab**
   Inner tabs for Data · Physics · Chemistry · DNA · Quantum.
2. **Pick a phenomenon**
   E.g. projectile motion; load the default values.
3. **Convert to maths**
   The phenomenon becomes an equation, vector, function or differential equation.
4. **Parameters**
   One key=value per line (v0=20, theta=45…).

Data Lab
Turns image, text, audio and video data into vectors, matrices, tensors, frequencies and embeddings.

Physics Lab
Motion, Newton, projectiles, harmonics, waves, electric field.

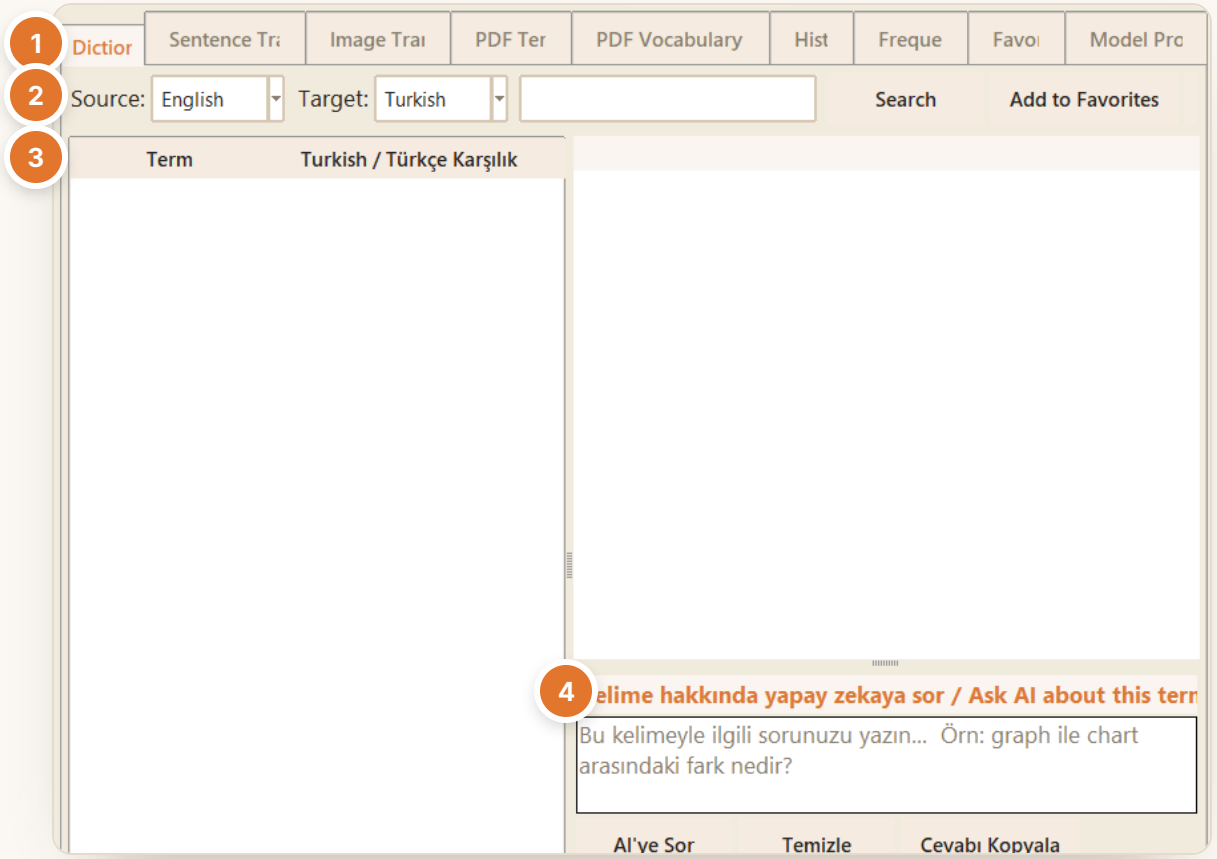
Chemistry Lab
Ideal gas, rate law, pH, decay, diffusion, buffers; periodic table, moles and stoichiometry.

DNA Lab
DNA ↔ maths: string, vector, k-mer, information; codon table; digital data → DNA.

Quantum
Qubits, gates, measurement, entanglement (Bell), QFT, Grover — in the language of vectors and matrices.

Note: the labs' interface labels are still in Turkish.

Language & Science Dictionary: get past the vocabulary barrier



1. **Nine sub-tabs**
   Dictionary, sentence and image translation, PDF terms, PDF vocabulary scanner, history, frequency, favourites, model profiles.
2. **Source → target language**
   Search English ↔ Turkish; add what you need to your favourites.
3. **Term list**
   Term and translation; select one to see its details alongside.
4. **Ask the AI about a word**
   “What is the difference between graph and chart?” — ask what a word means in maths.

- **PDF vocabulary scanner:** pulls the unfamiliar words out of a whole PDF at once.
- **Image/OCR translation:** reads and translates the text in a screenshot.

A scientific calculator and a tablet drawing board





1. **Keypad** — Fraction-aware (3/4 + 2/5 → an exact fraction); sqrt, log, sin…
2. **RAD / DEG · Explain** — Pick the angle unit; have the result explained.
3. **History & study note** — Every calculation is kept; usage tips alongside.
4. **Ask the AI** — “Add 15% VAT to 200”, “solve $x^2-5x+6=0$”.

5. **Drawing tools** — Pen, eraser, line, arrow, rectangle, circle, triangle, text; grid and axes.
6. **What did I draw? · Explain step by step** — Your drawing goes to the vision model and is interpreted — get your working checked.
7. **Send · Save** — Send it to the AI tutor or save it as PNG. Mouse, pen display or graphics tablet.

Shape the program around the way you study



The workspace strip: hide what you don't use with ×; manage all 13 workspaces from one list in the Workspace Manager.

Türkçe / English

The interface, the guide and the system instruction sent to the AI all switch to the language you choose.

Dark theme

A warm dark palette for night study (applied on restart). PDF pages stay light, like paper.

X Layout

Classic, mirrored or two independent PDF windows: put the lesson and the solution side by side.

Startup screen

Carry on from your last session (last PDF page and zoom included) or open straight on the Today dashboard.

User guide

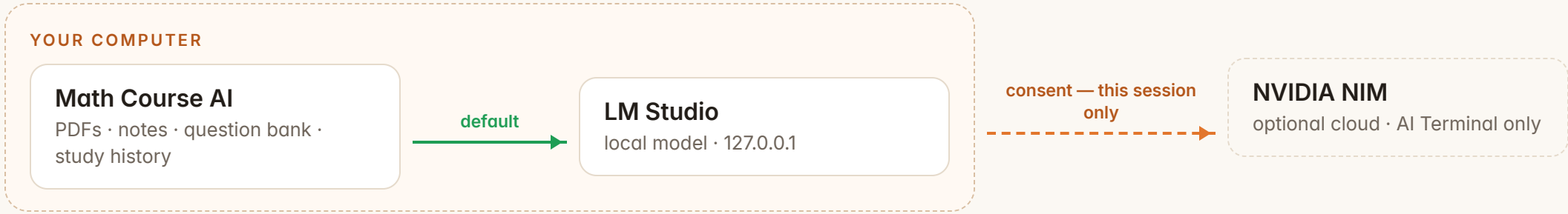
Settings → User Guide: 19 chapters inside the program, in Turkish and English.

Light start-up

Panels are built the first time you use them; a lab you never open never slows the program down.

Local by default — cloud only with your consent

Details are in PRIVACY.md and AI_TRANSPARENCY.md in the repository.



BY DEFAULT

- The AI is **local**: LM Studio runs on your own computer.
- Your study data (learner, answers, timings, notes) stays in a local SQLite database — on Windows under `%APPDATA%\MathCourseAI`.
- No account, no login, no server. No telemetry, no analytics, no update check.
- No course content is shipped: you bring your own material.

GOOD TO KNOW

- The cloud provider (NVIDIA NIM) is **optional**: it is only used if you pick it in the AI Terminal in that session and give consent. Then your prompt and the last 12 turns go to NVIDIA and are logged there.
- Code the AI writes never runs without your approval; in this release code execution is off by default.
- Deleting a learner erases every record belonging to them, exam answers included.
- AI answers can be wrong — always check them.

Up and running in three steps

1 · Download and run

Windows 10/11 (x64): Math Course AI-Windows.zip → extract to a folder → *Math Course AI.exe*.

macOS (Apple Silicon): Math Course AI-macOS.zip → extract → first launch: right-click → *Open* (not notarised).

Download: github.com/Azizsekerdil/math-course-ai/releases (v1.0.0)

From source: Python 3.11 · `pip install -r requirements.txt` · `python Math_Course_AI.pyw`

2 · The first 10 minutes

① Put your course PDFs into *Resources*, or pick a folder with Browse.
② Double-click a PDF in the tree.
③ Bank a question with Clip Question.
④ Start with Student Tracker → Exam.

No PDFs? Import a built-in generator from Question Packs and start straight away.

3 · Turn on the AI (optional)

① Install LM Studio from lmstudio.ai.
② Download a model (e.g. qwen2.5-math-7b-instruct).
③ Start the server in LM Studio: `127.0.0.1:1234`
④ Settings → model profiles, or “Auto Detect Loaded Models”.

The light in the top bar turns green.

If you get stuck: Settings → User Guide (19 chapters), chapter **Troubleshooting** · the **About** window shows your version and the shortcuts.

Math Course AI

Read · Annotate · Collect questions · Review · Ask when stuck — all in one window, with the AI on your own computer by default.

github.com/Azizsekerdil/math-course-ai | v1.0.0 · V48 | Source code: MIT | Beta · personal project

Who is it for?

Anyone following a maths course on their own, preparing for an exam, or wanting to review topics regularly.

What do you need?

Windows 10/11 or an Apple Silicon Mac and your own course PDFs — or the built-in question generators.

What it is not

It does not grade or certify you and ships no course content. Check the AI's answers.